Module A – Network Systems

IT NETWORK SYSTEMS ADMINISTRATION

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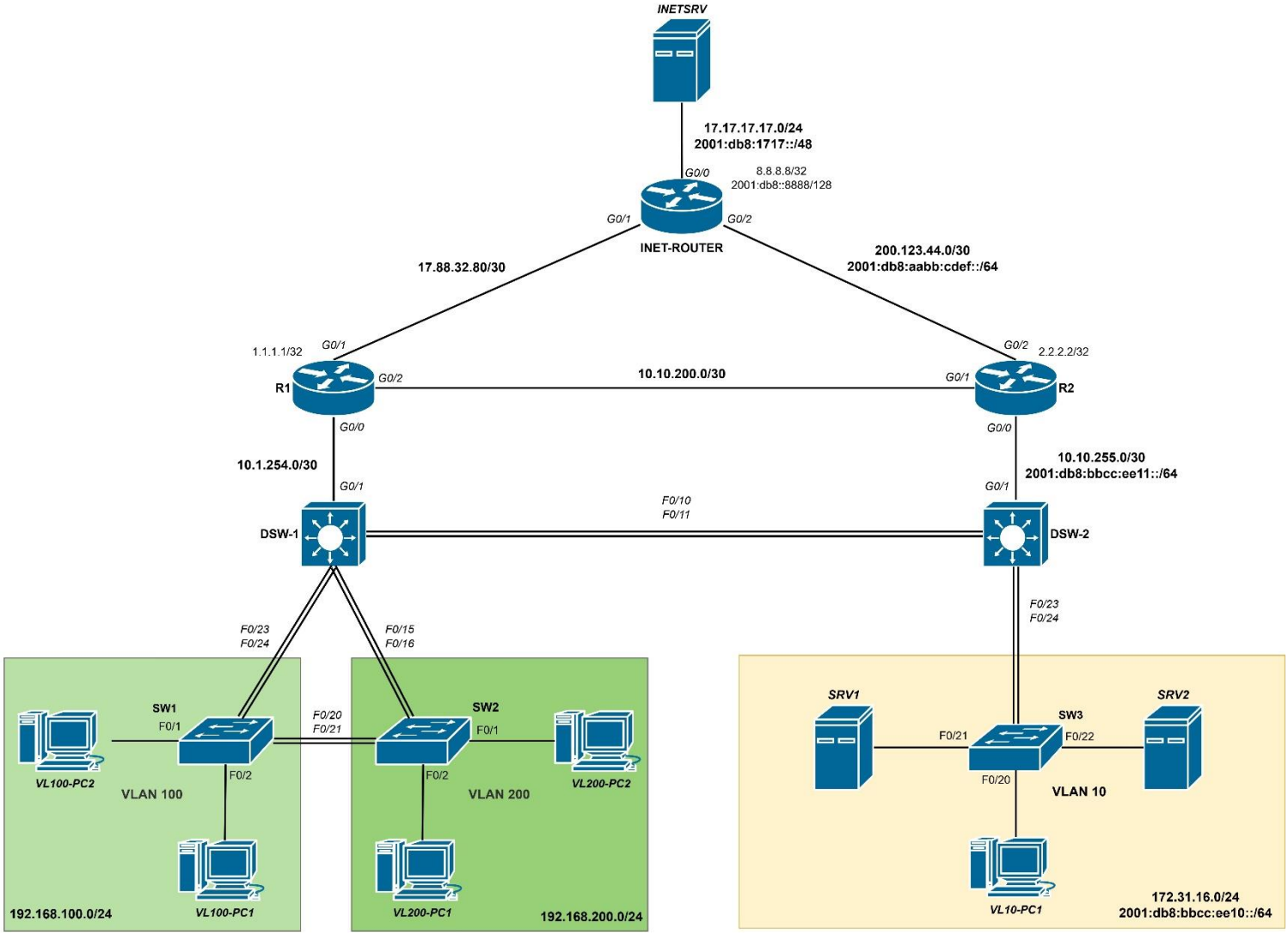
Topology



Basic Configuration

- Configure hostname on each network device refer to the topology
- Configure domain name on each network device to **skill39.local**
- Configure timezone on each network device to **WIB 7**
- Configure enable secret on each network device to **S3cret**

- Set IP address on each network device refer to the table below:

Device	Interface	IP Address
INET-ROUTER	Loopback0	8.8.8.8/32 2001:DB8::8888/128
	GigabitEthernet0/0	17.17.17.177/24 2001:DB8:1717::FFFF/48
	GigabitEthernet0/1	17.88.32.82/30
	GigabitEthernet0/2	200.123.44.2/30 2001:DB8:AABB:CDEF::1/64
R1	Loopback0	1.1.1.1/32
	GigabitEthernet0/0	10.1.254.2/30
	GigabitEthernet0/1	17.88.32.81/30
	GigabitEthernet0/2	10.10.200.1/30
R2	Loopback0	2.2.2.2/32
	GigabitEthernet0/0	10.10.255.2/30 2001:DB8:BBCC:EE11::FFFF/64
	GigabitEthernet0/1	10.10.200.2/30
	GigabitEthernet0/2	200.123.44.1/30 2001:DB8:AABB:CDEF::2/64
DSW-1	GigabitEthernet0/1	10.1.254.1/30
	Vlan100	192.168.100.254/24
	Vlan200	192.168.200.254/24
	Vlan999	192.168.99.1/29
DSW-2	GigabitEthernet0/1	10.10.255.1/30 2001:DB8:BBCC:EE11::1/64
	Vlan10	172.31.16.254/24 2001:DB8:BBCC:EE10::FFFF/64
	Vlan999	192.168.99.2/29
SW1	Vlan999	192.168.99.3/29
SW2	Vlan999	192.168.99.4/29
SW3	Vlan999	192.168.99.5/29

Switching Configuration

- Configure VTP version 2 on all switch
    - Set VTP domain to **skill39.local**
    - Set VTP password to **P@ssw0rd**
    - Make sure **DSW-1** configured as VTP Master
    - Other switches configured as VTP Client
- Create VLAN refer to the table below:

VLAN NUMBER	VLAN NAME
VLAN 10	VL-10
VLAN 100	VL-100
VLAN 200	VL-200
VLAN 999	NATIVE

- Configure port-channel refer to the list below:
    - Port-channel number **1** between **DSW-1** (*Fa0/23-24*) and **SW1** (*Fa0/23-24*)
        - Use **LACP** protocol
        - **DSW-1** configured as ***Active*** meanwhile **SW1** as ***Passive***
    - Port-channel **2** between **DSW-1** (*Fa0/15-16*) and **SW2** (*Fa0/15-16*)
        - Use **LACP** protocol
        - **DSW-1** configured as ***Active*** meanwhile **SW2** as ***Passive***
    - Port-channel **3** between **SW1** (*Fa0/20-21*) and **SW2** (*Fa0/20-21*)
        - Use **PAgP** protocol
        - Both switches must be set on **Desirable** mode
    - Port-channel number **6** between **DSW-2** (*Fa0/23-24*) and **SW-3** (*Fa0/23-24*)
        - Use **LACP** protocol
        - Both switches must be set on **Active** mode
    - Port-channel number **24** between **DSW-1** (*Fa0/10-11*) and **DSW-2** (*Fa0/10-11*)
        - Use **PAgP** protocol
        - Both switches must be set on **Desirable** mode
- Configure **interface between switches (*including Etherchannel port*) as trunk port** with native VLAN set to **999**
- On **SW1** configure interface Fa0/1-8 as **access port VLAN 100**
- On **SW2** configure interface Fa0/1-8 as **access port VLAN 200**
- On **SW3** configure interface Fa0/18-22 as **access port VLAN 10**

- Enable **port-fast feature** on access port (**SW1**, **SW2** and **SW3**)
- Enable `spanning-tree mode rapid-pvst` on all switches
- Set **DSW-1** as the **root bridge for VLAN 100, VLAN 200 and VLAN 999** and **DSW-2** as the **root bridge for VLAN 10**. Use lowest `priority` value to set switch as the Root Bridge!

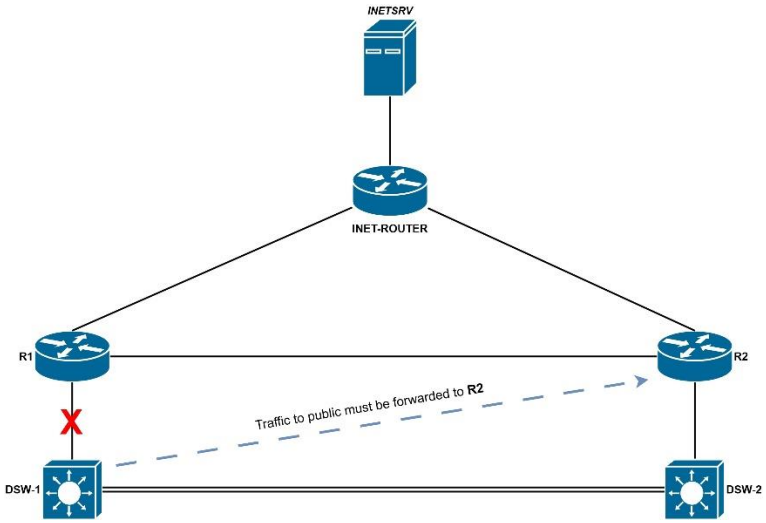
Routing Configuration

- Configure OSPF on **INET-ROUTER**, **R1** and **R2** using process-id **1** with specification below:
  - Configure `passive-interface default`
  - Only enable OSPF process on link that have **Public IP Address and link that only connected with Router**! Use default **OSPF area 0**
  - On **INET-ROUTER** advertise extra network below:
    - 17.17.17.0/24 area 0
    - 8.8.8.8/32 area 8
  - On **R1** advertise extra network below:
    - 1.1.1.1/32 area 1
  - On **R2** advertise extra network below:
    - 2.2.2.2/32 area 2
  - After adjacency established between routers, try to ping loopback IP like example below:

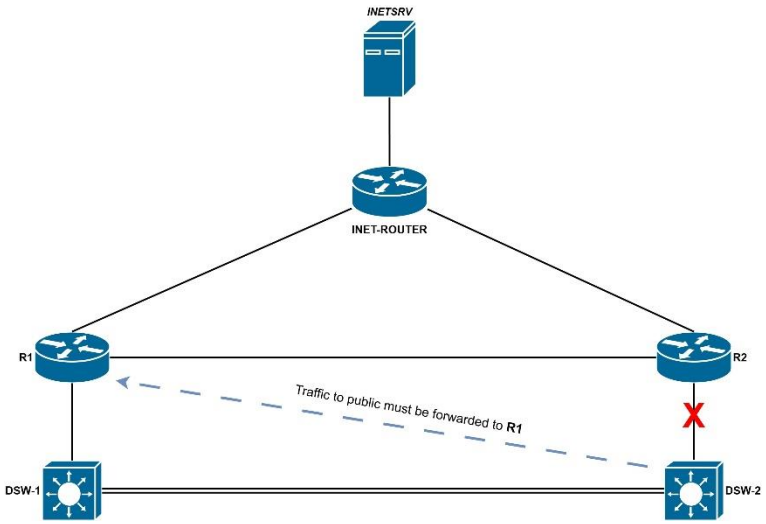
```
INET-ROUTER#ping 1.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

INET-ROUTER#ping 2.2.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

- Configure EIGRP on **DSW-1**, **DSW-2**, **R1** and **R2** using **AS number 10** with specification below:
  - Configure `passive-interface default`
  - Only enable EIGRP process on link that have **Private IP Address**. For **DSW** switches, establish adjacency on **VLAN 999**
  - Disable **auto summary**
  - Only advertise **Private IP Address** into EIGRP process

- Configure default static route on **DSW-1**, it must be forwarded to **R1**. Meanwhile **DSW-2** forwarded to **R2**
    - Add appropriate floating static route as the failover for default static route with **AD value 10** refer to scenario below:
        - If link between **R1** and **DSW-1** down, any traffic destined to public must be forwarded into **R2 (10.10.255.2**)



        - If link between **R2** and **DSW-2** down, any traffic destined to public must be forwarded into **R1 (10.1.254.2)**



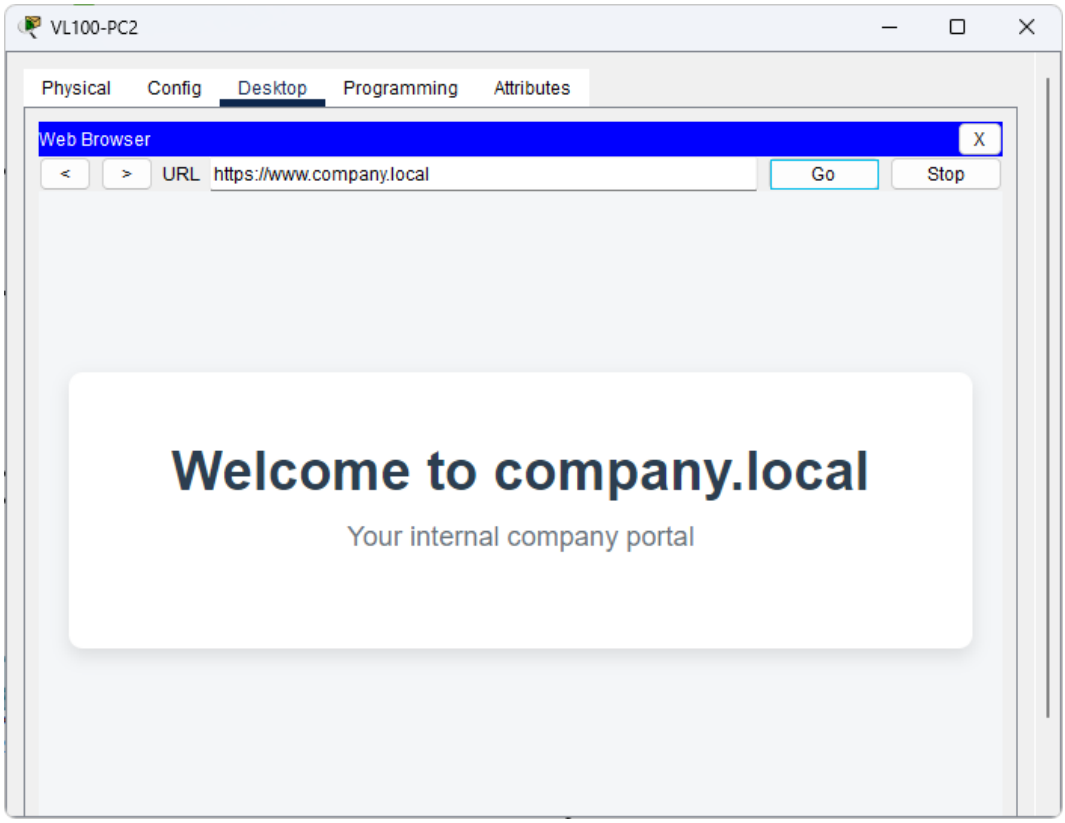
- Configure static IPv6 route on **DSW-2**, **R2** and **INET-ROUTER**. Make sure **SRV1 & SRV2** can communicate with **INETSRV** using IPv6

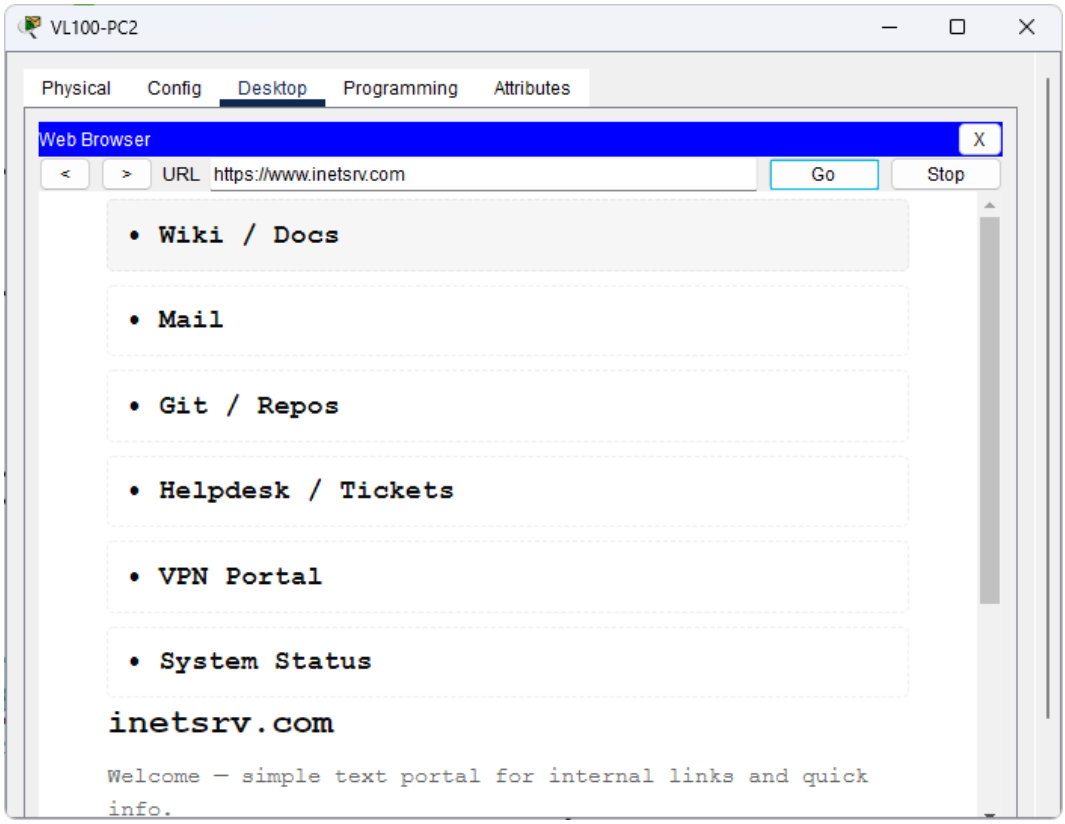
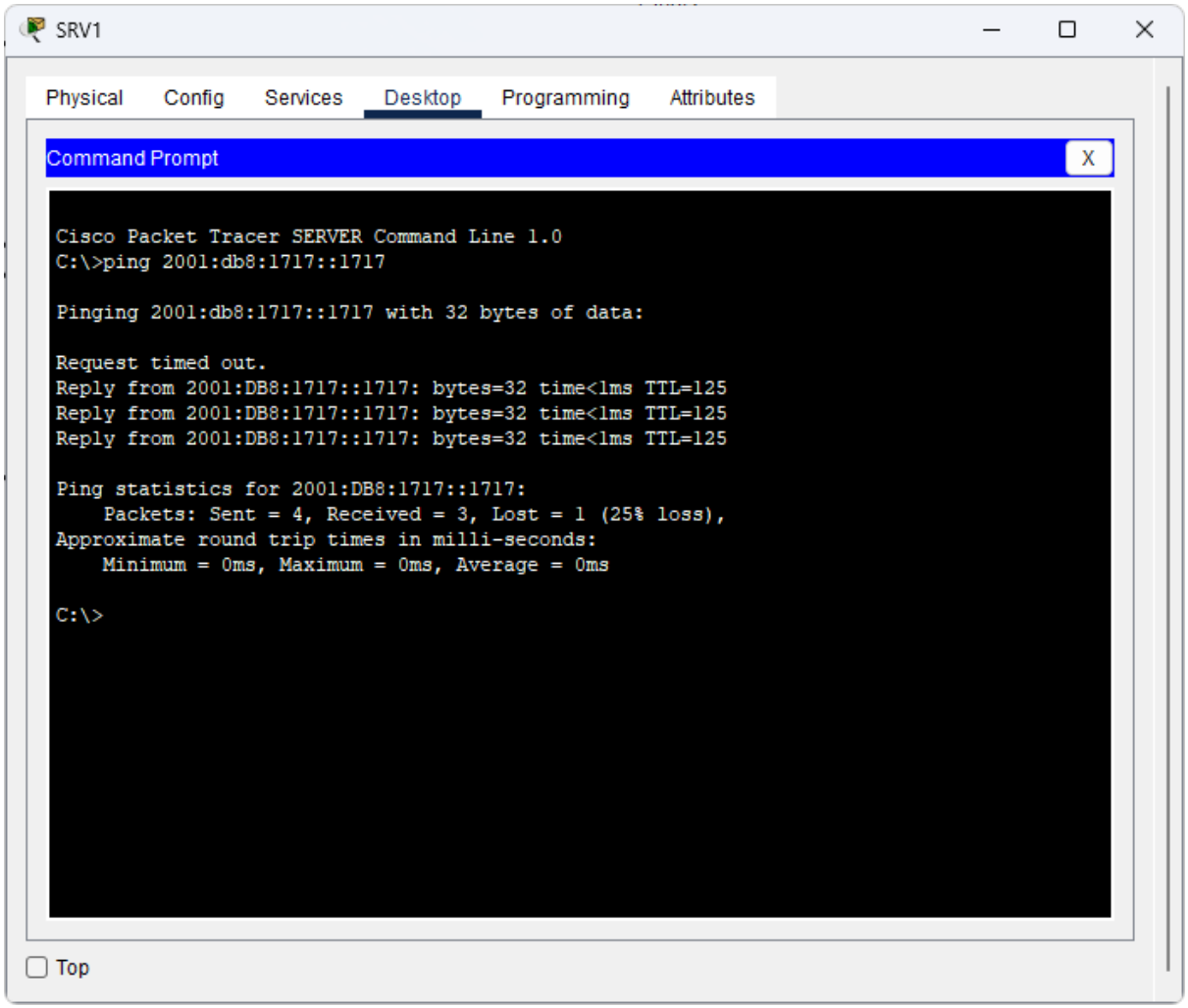
Service Configuration

- Configure DHCP pool on **DSW-1** refer to specification below:
    - DHCP for **VLAN 100**
        - Pool name: **VL100-POOL**
        - Network: **192.168.100.0/24**
        - Gateway: **192.168.100.254**
        - DNS: **172.31.16.10**
        - Excluded Address: **First 100 host address**
    - DHCP for **VLAN 200**
        - Pool name: **VL200-POOL**
        - Network: **192.168.200.0/24**
        - Gateway: **192.168.200.254**
        - DNS: **172.31.16.10**
        - Excluded Address: **First 100 host address**
- Configure DHCP pool on **DSW-2** refer for **VLAN 10** to specification below:
        - Pool name: **VL10-POOL**
        - Network: **172.31.16.0/24**
        - Gateway: **172.31.16.254**
        - DNS: **172.31.16.10**
        - Leased Address Range: **172.31.16.151 – 172.31.16.250**
- Configure **SRV2** (**172.31.16.20**) as the NTP server for **R1**, **R2**, **DSW-1** and **DSW-2**
- Configure SSH version 2 on **R1** and **R2**
    - Enable it on line vty0 – 11 and use username **ssh** with password **P@ssw0rd** for authentication
- Configure Port Address Translation on **R1** and **R2**
    - Use **access-list 1** that contains IP address below:
        - 192.168.0.0/16
        - 172.31.16.0/24
    - Configure appropriate **inside** and **outside** interface on routers

End goals:

PC can access **https://www.company.local** and **https://www.inetsrv.com**





SRV1 & SRV1 can communicate with **INETSRV** via IPv6 (**2001:db8:1717::1717**)